

Real-time scenario-based AWS EC2 Interview Questions & Answers:

This will cover **all aspects**: EC2 basics, instances, storage, networking, security, auto-scaling, monitoring, cost optimization, high availability, DevOps, troubleshooting, and real-world scenarios.

1–50: EC2 Basics & Core Concepts

1. **Q:** What is Amazon EC2?

A: EC2 is a web service that provides resizable compute capacity in the cloud, allowing users to launch virtual servers with different instance types, operating systems, and configurations.

2. **Q:** Scenario: You need a virtual server to host a web application for 1 year. Which EC2 pricing model is cost-efficient?

A: **Reserved Instances** for 1-year term to save costs compared to on-demand.

3. **Q:** Difference between EC2 and Lambda?

A: EC2 = full server control, long-running workloads. Lambda = serverless, short-duration functions with no server management.

4. **Q:** Scenario: You want ephemeral compute for batch jobs. Which instance type?

A: Spot Instances for cost savings.

5. **Q:** What are EC2 instance types?

A: General Purpose (t3/t4), Compute Optimized (c5), Memory Optimized (r5), Storage Optimized (i3), GPU Instances (p4/g4), etc.

6. **Q:** Scenario: Your app needs 4 vCPUs and 16GB RAM. Which instance type class?

A: General Purpose (e.g., **t3.xlarge**) or Compute Optimized if CPU-bound.

7. **Q:** Difference between on-demand, reserved, spot, and savings plan?

A: On-demand = pay per hour, no commitment. Reserved = 1–3 year commitment, cheaper. Spot = spare capacity at discount, can be interrupted. Savings plan = flexible discounted pricing for compute usage.

8. **Q:** Scenario: You need instances for dev/test environment that can stop/start frequently. Which pricing model?

A: **On-demand** for flexibility or **spot** if interruptions are acceptable.

9. **Q:** What is an Amazon Machine Image (AMI)?

A: A pre-configured template containing OS, application server, and apps to launch EC2 instances.

10. **Q:** Scenario: You need the same instance configuration in multiple regions. How?

A: Copy the AMI to target regions and launch instances.

11. **Q:** How do you access EC2 instances?

A: SSH for Linux, RDP for Windows.

12. **Q:** Scenario: You lost your private key for SSH. How to access Linux EC2?
A: Stop instance, detach root volume, attach to another instance, update ~/.ssh/authorized_keys, reattach, start instance.
13. **Q:** What are instance states?
A: Pending, Running, Stopping, Stopped, Shutting-down, Terminated.
14. **Q:** Scenario: Your EC2 instance is stopped but billing continues. Why?
A: EBS volume attached and provisioned Elastic IP incurs charges.
15. **Q:** What are Elastic IPs?
A: Static public IPv4 addresses for EC2, which can be remapped to other instances.
16. **Q:** Scenario: Elastic IP attached but not in use. Any cost implications?
A: Yes, AWS charges for unattached Elastic IPs.
17. **Q:** Difference between instance store and EBS?
A: Instance store = ephemeral storage tied to instance lifecycle. EBS = persistent block storage.
18. **Q:** Scenario: Instance store lost data after reboot. Why?
A: Instance store is ephemeral; data is lost when instance stops or terminates.
19. **Q:** What is EC2 User Data?
A: Script executed during instance launch to configure the instance automatically.
20. **Q:** Scenario: You need auto-installation of Apache/Nginx on instance launch. How?
A: Add shell commands in **User Data** script.
21. **Q:** What is EC2 metadata service?
A: Provides instance info (instance-id, security groups, AMI ID) accessible via HTTP endpoint.
22. **Q:** Scenario: You want dynamic configuration from instance itself. How?
A: Use EC2 Instance Metadata Service (IMDS v2 recommended).
23. **Q:** What is an instance profile?
A: Attaches IAM role to EC2 instance for AWS API access.
24. **Q:** Scenario: EC2 needs to read/write S3 bucket securely. How?
A: Attach IAM role via instance profile with S3 permissions.
25. **Q:** Can an EC2 instance have multiple ENIs?
A: Yes, supported depending on instance type.
26. **Q:** Scenario: You need private and public network interfaces on same instance. How?
A: Attach multiple ENIs and configure routing/security groups.
27. **Q:** What is a placement group?
A: A logical grouping to influence instance placement (clustered, spread, partition) for performance/HA.

28. **Q:** Scenario: Low-latency HPC workload. Which placement strategy?
 A: Cluster placement group.
29. **Q:** Difference between Cluster, Spread, and Partition placement groups?
 A: Cluster = high throughput/low latency. Spread = isolate instances across hosts for failure domain. Partition = divides instances into logical partitions for HA.
30. **Q:** Scenario: Need fault-tolerant architecture across AZs. Which placement group?
 A: Spread or Partition placement group.
31. **Q:** Can you change EC2 instance type after launch?
 A: Yes, stop instance → modify instance type → start.
32. **Q:** Scenario: You need more CPU for few hours. How to scale?
 A: Resize instance (vertical scaling) or use Auto Scaling group with launch template.
33. **Q:** What is EC2 Auto Scaling?
 A: Automatically adjusts number of instances based on demand using scaling policies.
34. **Q:** Scenario: Traffic spikes 10x daily at 6 PM. How to manage cost/performance?
 A: Use Auto Scaling scheduled policies to scale out/in at 6 PM and off-peak.
35. **Q:** Difference between horizontal and vertical scaling?
 A: Horizontal = add/remove instances. Vertical = change instance type.
36. **Q:** What is EC2 launch template?
 A: Template containing instance configuration (AMI, instance type, key pair, security groups) for launching instances.
37. **Q:** Scenario: Multiple ASGs require same configuration. Solution?
 A: Use **Launch Template** across ASGs.
38. **Q:** Can you attach EBS volume to running EC2?
 A: Yes, online attach/detach supported.
39. **Q:** Scenario: Need increased storage for Linux EC2 without downtime. How?
 A: Use EBS **elastic volume modification**, extend filesystem online.
40. **Q:** Can you encrypt EBS volumes?
 A: Yes, using KMS-managed keys.
41. **Q:** Scenario: Need encryption for existing unencrypted volume. How?
 A: Take snapshot → copy with encryption enabled → create new volume → attach.
42. **Q:** Difference between instance store-backed vs EBS-backed instance?
 A: Instance store-backed loses data on stop. EBS-backed persists.
43. **Q:** What is EC2 hibernation?
 A: Saves instance RAM to EBS and stops instance; resumes quickly with OS state intact.

44. **Q:** Scenario: Long-running process needs paused state. Solution?
A: Enable EC2 hibernation.
45. **Q:** Can you attach multiple EBS volumes to same instance?
A: Yes, limited by instance type ENI and volume count limits.
46. **Q:** How to monitor EC2 health?
A: Use **CloudWatch metrics**, **Status Checks**, and **CloudTrail**.
47. **Q:** Scenario: Instance failed System Status Check. Next steps?
A: Stop/start instance, check underlying hardware, contact AWS if persistent.
48. **Q:** Scenario: Instance failed Instance Status Check. Fix?
A: Check OS boot logs, disk issues, SSH connectivity, kernel panic.
49. **Q:** What is EC2 placement tenancy?
A: Default (shared hardware) vs Dedicated host (physical server fully for you).
50. **Q:** Scenario: Compliance requires dedicated physical host. Solution?
A: Use **Dedicated Host** tenancy.

51–100: Networking & Security

51. **Q:** How does EC2 connect to VPC?
A: EC2 is launched in a subnet within VPC; connectivity controlled via route tables, internet gateway, NAT, and security groups.
52. **Q:** Scenario: EC2 needs private subnet access to internet. How?
A: Use NAT gateway/instance in public subnet.
53. **Q:** Difference between Security Groups and NACLs?
A: Security Group = instance-level firewall (stateful). NACL = subnet-level firewall (stateless).
54. **Q:** Scenario: All traffic blocked despite security group open. Why?
A: Check NACL rules; may be denying traffic.
55. **Q:** What is Elastic Load Balancer (ELB) with EC2?
A: Distributes incoming traffic across multiple EC2 instances.
56. **Q:** Scenario: Need high availability for web app. Solution?
A: Use EC2 instances across AZs behind **ALB**.
57. **Q:** Difference between ALB, NLB, CLB?
A: ALB = HTTP/HTTPS, path-based routing. NLB = TCP, high performance. CLB = legacy.
58. **Q:** Scenario: Need SSL termination at load balancer. Which ELB?
A: Application Load Balancer.

59. Q: What is VPC endpoint for EC2?

A: Private connection to AWS services without internet access.

60. Q: Scenario: EC2 must access S3 without public IP. Solution?

A: Use **S3 VPC Endpoint (Gateway type)**.

61. Q: How to implement security patching on EC2?

A: Use Systems Manager Patch Manager or manual updates via SSH/SSM.

62. Q: Scenario: Need to patch hundreds of EC2 Linux instances automatically. Solution?

A: Use **AWS Systems Manager Patch Manager** with patch baselines.

63. Q: How to enable MFA for EC2 access?

A: IAM policies + MFA enforced on AWS console/API; EC2 SSH uses key pair.

64. Q: What is EC2 key pair?

A: Public-private key pair for SSH/RDP authentication.

65. Q: Scenario: Lost private key for production instance. Fix?

A: Stop instance → detach volume → attach to another instance → add new key to authorized_keys.

66. Q: How to restrict EC2 API access?

A: Use IAM roles, policies, and VPC endpoint policies.

67. Q: Scenario: EC2 instance compromised. Immediate action?

A: Isolate in security group, snapshot volume, analyze, rotate keys/credentials.

68. Q: Difference between default VPC and custom VPC?

A: Default VPC = preconfigured with public subnets. Custom = fully configurable.

69. Q: Scenario: Multi-tier architecture with public web, private app, DB subnet. How?

A: Public subnet → ALB + web EC2

. Private subnet → app EC2 + DB RDS. NAT for outbound internet from private subnet.

70. Q: How to enforce HTTPS only on EC2?

A: Configure Security Groups to allow only 443; optionally enforce via ALB listener rules.

71. Q: How to limit SSH access to specific IPs?

A: Security Group inbound rule with specific CIDR.

72. Q: Scenario: SSH from dynamic ISP IPs. Solution?

A: Use **bastion host** with dynamic IP allowed, or VPN connection to VPC.

73. Q: What is EC2 instance metadata service v2?

A: Session-oriented API for instance metadata with IMDSv2 for security.

74. Q: Scenario: Prevent SSRF attacks exploiting metadata service. Solution?

A: Enforce IMDSv2 requirement.

75. **Q:** How to enable private DNS hostname for EC2?
A: Enable enableDnsHostnames and enableDnsSupport in VPC.
76. **Q:** Scenario: EC2 cannot resolve DNS. Fix?
A: Check VPC DNS settings, DHCP options, security group egress rules.
77. **Q:** Difference between public and private IP on EC2?
A: Public IP = internet routable. Private IP = internal VPC routing only.
78. **Q:** Scenario: EC2 in private subnet needs RDP to Windows. How?
A: Use bastion host or Systems Manager Session Manager.
79. **Q:** What is source/destination check?
A: Enabled by default; prevents instance from routing traffic not destined for itself.
80. **Q:** Scenario: NAT instance cannot route traffic. Fix?
A: Disable source/destination check.
81. **Q:** How to attach multiple IP addresses to EC2?
A: Use multiple private IPs via secondary ENI.
82. **Q:** Scenario: Multi-homed EC2 for hybrid cloud. Approach?
A: Attach multiple ENIs for public/private connections.
83. **Q:** How to implement EC2 security compliance auditing?
A: Use Config rules, CloudTrail, and Security Hub.
84. **Q:** Scenario: Detect unauthorized AMI launches. How?
A: CloudTrail + Config rules on AMI creation.
85. **Q:** Difference between Security Group stateful vs NACL stateless?
A: Security Group allows response traffic automatically. NACL must explicitly allow return traffic.
86. **Q:** Scenario: Web server in public subnet but cannot serve clients. Check?
A: SG inbound, route table, subnet ACL, internet gateway attachment.
87. **Q:** Scenario: Windows EC2 cannot RDP. Likely reasons?
A: Security group inbound 3389, Network ACL, firewall, or RDP service disabled.
88. **Q:** How to configure bastion host securely?
A: Public subnet, hardened AMI, limited SSH access, CloudWatch logging.
89. **Q:** Scenario: Bastion host compromised. Steps?
A: Isolate, snapshot volume, rotate keys, review CloudTrail, redeploy hardened host.
90. **Q:** How to restrict EC2 internet access?
A: Remove IGW/NAT, security group rules, NACLs, or use private subnet.

91. Q: What is Security Group reference?

A: Allows SG to reference another SG as source/destination.

92. Q: Scenario: App server allowed only DB server to connect. How?

A: Security group referencing DB server SG.

93. Q: Can EC2 instances in different regions communicate privately?

A: No direct private IP. Use VPN/Transit Gateway.

94. Q: What is EC2 placement group impact on networking?

A: Cluster PG → low-latency, high-throughput. Spread PG → isolated.

95. Q: Scenario: EC2 network throughput low. Check?

A: Instance type network limits, ENI attachment, placement group.

96. Q: What is ENI?

A: Elastic Network Interface, virtual NIC attached to EC2.

97. Q: Scenario: High availability web servers. Solution?

A: Multi-AZ deployment, ALB, Auto Scaling.

98. Q: Scenario: Need private EC2 to access S3 with no IGW. How?

A: Use S3 VPC Endpoint.

99. Q: Scenario: NAT gateway fails. What happens?

A: Private subnet loses internet connectivity.

100. Q: Difference between NAT instance and NAT gateway?

A: NAT gateway = managed, scalable, redundant. NAT instance = self-managed, less scalable.

101–150: Storage – EBS, Instance Store, EFS, S3 Integration

101. Q: Difference between instance store and EBS?

A: Instance store = ephemeral, tied to instance lifecycle; EBS = persistent block storage.

102. Q: Scenario: EC2 rebooted, instance store data lost. Why?

A: Instance store is ephemeral; data does not persist across stop/start.

103. Q: Can you attach EBS to running EC2?

A: Yes, online attach/detach supported.

104. Q: Scenario: Need to increase EBS volume size without downtime. How?

A: Use **Elastic Volume Modification** → extend filesystem.

105. Q: Can EBS volumes be encrypted?

A: Yes, using AWS KMS-managed keys.

106. Q: Scenario: Existing unencrypted EBS. How to encrypt?

A: Take snapshot → copy with encryption enabled → create new volume → attach.

107. Q: What is EBS volume types?

A: gp3/gp2 (General Purpose SSD), io2/io1 (Provisioned IOPS SSD), st1 (Throughput Optimized HDD), sc1 (Cold HDD).

108. Q: Scenario: High IOPS database workload. Which volume type?

A: io2/io1 with provisioned IOPS.

109. Q: How does EBS snapshot work?

A: Incremental backup stored in S3.

110. Q: Scenario: Need cross-region backup. Solution?

A: Copy EBS snapshot to target region.

111. Q: What is EFS?

A: Elastic File System – fully managed, scalable NFS file system for multiple EC2 access.

112. Q: Scenario: Multiple EC2 instances need shared filesystem. Which service?

A: **EFS** with NFS mount.

113. Q: Difference between EBS and EFS?

A: EBS = block storage, one instance; EFS = file storage, multiple instances concurrently.

114. Q: Scenario: EC2 instance needs temporary storage. Which option?

A: Instance store for ephemeral fast storage.

115. Q: Can EBS volumes be attached to multiple instances?

A: No, except **EBS Multi-Attach** supported for io1/io2 volumes in Linux.

116. Q: Scenario: EC2 in multi-AZ with shared volume. Solution?

A: Use **EFS** or FSx for shared access, EBS Multi-Attach in same AZ.

117. Q: How to monitor EBS performance?

A: CloudWatch metrics: VolumeReadOps, VolumeWriteOps, VolumeReadBytes, VolumeWriteBytes, BurstBalance.

118. Q: Scenario: VolumeWriteOps high, instance slow. Solution?

A: Increase volume type/IOPS, check instance network limits.

119. Q: How to resize EFS storage?

A: Automatic scaling; no manual resizing needed.

120. Q: Scenario: EC2 cannot mount EFS. Check?

A: Security group allows NFS (2049), correct VPC/subnet, mount target exists.

121–150: Storage – EBS/EFS, S3, Snapshots

121. Q: Scenario: EC2 root volume running out of space. How to extend?

A: Use **Elastic Volume Modification** → extend EBS volume → resize filesystem (xfs_growfs or resize2fs).

122. Q: Scenario: EBS snapshot restores slowly. Solution?

A: Enable **Fast Snapshot Restore** for region-specific instant availability.

123. Q: EC2 needs high-throughput storage for big data. Options?

A: Use **st1 (Throughput Optimized HDD)** or **EFS** depending on shared access needs.

124. Q: Scenario: Multi-instance application requires shared file system. How?

A: Mount **EFS** with NFS across instances.

125. Q: Can EBS volumes be resized while attached?

A: Yes, online resizing supported; extend filesystem afterward.

126. Q: Scenario: Instance requires both high IOPS and encryption. Which volume type?

A: **io2/io2 Block Express** with KMS encryption.

127. Q: EC2 needs cross-region backup for compliance. How?

A: Copy EBS snapshots or AMIs to target region.

128. Q: Scenario: EBS volume attached to wrong instance. Solution?

A: Detach, attach to correct instance; ensure OS recognizes device.

129. Q: What is EBS Multi-Attach?

A: io1/io2 volumes can be attached to multiple Linux instances in the same AZ for shared access.

130. Q: Scenario: Application requires read-only shared storage across AZs. How?

A: Use **S3** or **EFS** (multi-AZ support).

131. Q: How to monitor EBS performance?

A: CloudWatch metrics: VolumeReadOps, VolumeWriteOps, VolumeThroughputPercentage, BurstBalance.

132. Q: Scenario: VolumeWriteOps high, performance lag. Solution?

A: Increase IOPS, upgrade instance type, check network limits.

133. Q: EC2 requires backup of ephemeral data. How?

A: Copy data to **EBS snapshot** or **S3**, as instance store is ephemeral.

134. Q: Scenario: Need encrypted EBS after instance launch. How?

A: Snapshot existing volume → copy snapshot with encryption → create encrypted volume → attach.

135. Q: What is difference between gp2 and gp3 EBS volumes?

A: gp3 = provisioned IOPS & throughput independently; cheaper & better performance flexibility.

136. Q: Scenario: Multiple EC2s need access to shared file system. Solution?

A: Use **EFS** with NFS mounts; scalable & concurrent access.

137. Q: EC2 needs high IOPS for database. Which volume type?

A: **io2 Block Express** for sub-millisecond latency.

138. Q: Scenario: EBS snapshot costs rising. Solution?

A: Delete old snapshots, automate with **Lifecycle Manager** policies.

139. Q: EC2 ephemeral storage lost on stop. Solution for persistence?

A: Migrate to **EBS** or **EFS** depending on shared access requirements.

140. Q: Scenario: Need to restore instance quickly in DR. Solution?

A: Use **AMI + snapshot** → launch new EC2 quickly.

141. Q: EC2 needs cross-AZ shared storage. Option?

A: **EFS** with mount targets in each AZ.

142. Q: Scenario: Large dataset read-heavy, multiple instances. Which storage?

A: **EFS** (for shared read), **S3** (for object-based), or combination.

143. Q: How to secure EBS volumes?

A: Enable encryption (**KMS**), **IAM** roles for access, snapshots encrypted.

144. Q: Scenario: Volume attached but not recognized by OS. Fix?

A: Rescan devices, check **/dev/** path, mount filesystem.

145. Q: How to migrate instance store data to persistent storage?

A: Copy files to **EBS** volume or **S3** before instance termination.

146. Q: Scenario: EC2 needs high throughput temporary storage. Option?

A: **Instance store** for ephemeral fast storage.

147. Q: EC2 EBS volume detached accidentally. How to recover?

A: Reattach; ensure filesystem integrity and mount properly.

148. Q: Scenario: EC2 root volume corrupted. How to recover?

A: Detach volume, attach to rescue instance, repair filesystem, reattach.

149. Q: EBS snapshot can restore to larger volume?

A: Yes, volume size can be increased during restore.

150. Q: Scenario: EC2 requires daily automated backup. How?

A: Use **EBS snapshot with Lifecycle Manager** or CloudWatch scheduled Lambda scripts.

151–200: Auto Scaling & High Availability

151. Q: What is EC2 Auto Scaling?

A: Automatically adjusts number of EC2 instances based on load.

152. Q: Scenario: Traffic spikes at 6 PM daily. How to scale?

A: Use **scheduled scaling policies** in Auto Scaling Group (ASG).

153. Q: Difference between horizontal and vertical scaling?

A: Horizontal = add/remove instances; Vertical = change instance type.

154. Q: Scenario: Need to scale memory-heavy workload. Approach?

A: Vertical scaling: change to higher memory instance.

155. Q: How does ASG detect unhealthy instance?

A: Health checks (EC2 status checks, ELB health checks).

156. Q: Scenario: ASG not replacing failed instance. Check?

A: Ensure health check type and grace period configured correctly.

157. Q: What is launch configuration vs launch template?

A: Launch configuration = static template; Launch template = versioned, flexible, supports multiple ASGs.

158. Q: Scenario: Multiple ASGs with same config. Solution?

A: Use **Launch Template**.

159. Q: How to implement Multi-AZ high availability?

A: Deploy instances across multiple AZs, use ELB.

160. Q: Scenario: Single AZ fails, app downtime. Fix?

A: Deploy Multi-AZ ASG with cross-AZ load balancer.

161. Q: What is ELB health check?

A: Periodically pings instance endpoint; unhealthy instances removed from load balancer.

162. Q: Scenario: ELB keeps removing instance. Check?

A: Security group, response codes, app listening port.

163. Q: Difference between AZ and region?

A: Region = geographical area; AZ = isolated datacenter within region.

164. Q: Scenario: Cross-region disaster recovery. How to deploy EC2?

A: Replicate AMI, data, and ASG in secondary region; optionally use Route53 for failover.

165. Q: How to handle sticky sessions?

A: Enable **ELB session stickiness** or use client-side cookies.

166. Q: Scenario: Need minimum 5 instances always running. How?

A: Set **ASG MinSize=5**, DesiredCapacity=5.

167. Q: Scenario: Reduce over-provisioning at night. How?

A: Use **scheduled scaling policies** or target tracking with metrics.

168. Q: What is target tracking scaling policy?

A: Auto scale instances to maintain metric (e.g., CPUUtilization) near target.

169. Q: Scenario: App CPU < 20% but ASG not scaling down. Check?

A: Cooldown period may prevent immediate scale-in.

170. Q: Difference between ASG scaling policies: simple, step, target tracking?

A: Simple = one step, step = multiple step thresholds, target tracking = automatic based on metric.

171–200: Auto Scaling & High Availability

171. Q: Scenario: Auto Scaling not triggering. Check?

A: Verify CloudWatch metric, scaling policy, cooldown period, min/max instances.

172. Q: ASG scale-in fails. Possible reasons?

A: Lifecycle hooks delaying termination, instance in service with ELB, cooldown period.

173. Q: Scenario: Scheduled scale-out for batch jobs. Approach?

A: Configure **scheduled scaling policy** with exact time and instance count.

174. Q: ASG instance unhealthy but not replaced. Why?

A: Incorrect health check type (EC2 vs ELB), grace period too long.

175. Q: Scenario: Maintain 3 instances minimum. How?

A: ASG **MinSize=3**, DesiredCapacity=3.

176. Q: How to implement Multi-AZ ASG?

A: Launch instances across multiple AZs; ELB distributes traffic.

177. Q: Scenario: ELB removes healthy instance. Why?

A: Health check misconfiguration, app listening port, SG restrictions.

178. Q: Difference between step and target tracking policy?

A: Step = different actions for thresholds; target tracking = maintain metric at target.

179. Q: Scenario: Scale down not reducing cost. Why?

A: MinSize too high, ASG cooldown, metric thresholds not met.

180. Q: Scheduled vs dynamic scaling?

A: Scheduled = known time events; dynamic = real-time metric-based.

181. Q: Scenario: CPU < 20% but instances not terminated. Solution?
A: Check **cooldown period**, metric evaluation period, step scaling thresholds.
182. Q: ASG cross-region support?
A: No; ASG operates per region; use multiple ASGs for cross-region DR.
183. Q: Scenario: Batch processing spikes CPU. How to handle?
A: Use **scheduled or target-tracking ASG** to scale out during batch.
184. Q: ASG minimum instances for SLA?
A: Set **MinSize** to maintain required availability.
185. Q: Scenario: AZ failure. How ASG maintains availability?
A: Instances launched in other AZs automatically to meet desired capacity.
186. Q: Health check types in ASG?
A: EC2 (instance running), ELB (application responsiveness).
187. Q: Scenario: ASG not replacing unhealthy instance. Check?
A: Health check configuration, ELB attachment, termination policies.
188. Q: Lifecycle hooks purpose?
A: Pause scale-in/out to perform custom actions (drain connections, backup).
189. Q: Scenario: Need zero-downtime deployment with ASG. How?
A: Use **Blue/Green deployment**, replace instances gradually.
190. Q: ASG integration with CloudWatch?
A: Metrics trigger scaling policies (CPU, Network, custom metrics).
191. Q: Scenario: ASG scaling too slow. How to improve?
A: Reduce cooldown period, use step policies, adjust evaluation periods.
192. Q: ASG scaling limits?
A: MaxSize for upper limit; MinSize for lower limit.
193. Q: Scenario: Multi-AZ EC2 HA with ASG & ELB. How?
A: Deploy instances across AZs, ELB distributes traffic; ASG maintains capacity.
194. Q: Target tracking policy example?
A: Maintain average CPU at 50% → ASG scales up/down automatically.
195. Q: Scenario: ASG scales incorrectly during low load. Check?
A: Verify CloudWatch metric, evaluation period, step scaling thresholds.
196. Q: Spot instances in ASG. Scenario?
A: Configure **Mixed Instances Policy** → on-demand base, spot for bursts.

197. Q: Scenario: ASG replacing healthy instance repeatedly. Fix?

A: Health check misconfigured; check termination policies.

198. Q: How to test ASG scaling policies?

A: Simulate load via **stress testing** or CloudWatch metrics manipulation.

199. Q: Scenario: Need consistent AZ distribution for instances. How?

A: Set **availability zones** in ASG configuration.

200. Q: Scenario: ASG not scaling due to metric delay. Fix?

A: Enable detailed monitoring (1-min granularity).

201–250: Monitoring & CloudWatch

201. Q: How to monitor EC2 performance?

A: CloudWatch metrics: CPU, NetworkIn/Out, DiskRead/Write, StatusChecks, Memory (custom).

202. Q: Scenario: Instance CPU high but CloudWatch normal. Why?

A: Memory/IO bottleneck not visible in default metrics; enable detailed monitoring or custom metrics.

203. Q: What are CloudWatch Alarms?

A: Notifications triggered when metric crosses threshold.

204. Q: Scenario: Receive repeated alarms for same instance. How to reduce?

A: Adjust evaluation periods, enable OK state suppression, tune thresholds.

205. Q: What is CloudWatch Logs?

A: Centralized logging for EC2 (application/system logs).

206. Q: Scenario: Application log monitoring for errors. Solution?

A: Install CloudWatch Agent, push logs to CloudWatch Logs.

207. Q: How to monitor ELB + EC2 metrics?

A: CloudWatch metrics: RequestCount, Latency, HealthyHostCount.

208. Q: Scenario: Latency spikes on EC2. How to investigate?

A: Check CloudWatch CPU/Memory/Network, ELB metrics, application logs.

209. Q: Difference between basic and detailed monitoring?

A: Basic = 5-min intervals, Detailed = 1-min intervals (cost extra).

210. Q: Scenario: You need 1-min granularity for auto-scaling. Solution?

A: Enable **detailed monitoring**.

211–250: Monitoring & CloudWatch

211. Q: Scenario: Need memory monitoring for EC2. How?

A: Enable **CloudWatch Agent** and push custom memory metrics.

212. Q: EC2 network performance metrics in CloudWatch?

A: NetworkIn, NetworkOut, NetworkPacketsIn, NetworkPacketsOut.

213. Q: Scenario: Instance shows high CPU but app slow. Check?

A: Memory/IO metrics, CloudWatch Logs, Disk/Network throughput.

214. Q: CloudWatch alarm triggers EC2 reboot. How?

A: Alarm → SNS → Lambda → Systems Manager or API call to reboot instance.

215. Q: Scenario: ELB latency high. Investigate?

A: Check EC2 CPU/memory, network metrics, application logs, placement group.

216. Q: Detailed vs basic monitoring?

A: Basic = 5-min; Detailed = 1-min intervals.

217. Q: Scenario: Auto Scaling requires 1-min metrics. Solution?

A: Enable **detailed monitoring** on EC2.

218. Q: How to aggregate CloudWatch logs across multiple EC2s?

A: Use centralized CloudWatch Logs group with subscriptions.

219. Q: Scenario: Alarm flapping frequently. Fix?

A: Adjust threshold, evaluation periods, or use anomaly detection.

220. Q: CloudWatch custom metrics example?

A: App response time, queue length, memory utilization.

221. Q: Scenario: EC2 app crashes silently. How to debug?

A: CloudWatch Logs, X-Ray tracing, detailed monitoring.

222. Q: Scenario: Disk IO bottleneck detected. Fix?

A: Upgrade EBS volume, enable EBS-optimized instance, check instance type.

223. Q: EC2 CPU credit exhaustion scenario. Solution?

A: Burstable T2/T3 instance, monitor CPUCreditBalance, consider larger instance.

224. Q: Scenario: CloudWatch alarm triggers at wrong time. Fix?

A: Adjust evaluation period, threshold, or metric aggregation.

225. Q: EC2 application latency monitoring. Tool?

A: CloudWatch metrics, CloudWatch Logs, X-Ray traces.

226. Q: Scenario: Need historical metric analysis. Solution?
A: CloudWatch metrics history, export to S3, use Athena/QuickSight.
227. Q: Scenario: EC2 auto-recovery trigger fails. How to investigate?
A: Check CloudWatch Alarm, IAM permissions, Auto Recovery enabled.
228. Q: Scenario: Multiple EC2 metrics need centralized alerting. Solution?
A: CloudWatch dashboard + alarms + SNS notifications.
229. Q: Scenario: Instance health checks failing intermittently. Fix?
A: Analyze system logs, ELB health, network config, instance metrics.
230. Q: CloudWatch Logs retention setting?
A: Configure per log group; delete after N days to save costs.
231. Q: Scenario: Need aggregated metrics across AZs. How?
A: CloudWatch metric math, or consolidate via CloudWatch dashboards.
232. Q: Scenario: Need notification if EC2 disk >80% used.
A: CloudWatch Agent → custom disk metric → Alarm → SNS.
233. Q: Scenario: Metrics delayed due to high load. Fix?
A: Enable detailed monitoring, check CloudWatch quota.
234. Q: CloudWatch anomaly detection use case?
A: Detect unusual CPU, memory, or network patterns automatically.
235. Q: Scenario: EC2 stops sending metrics after reboot. Fix?
A: CloudWatch Agent must start on boot; check IAM role.
236. Q: Scenario: Custom app logs not appearing. Fix?
A: Verify CloudWatch Logs agent config, permissions, path.
237. Q: Scenario: EC2 auto-scale not triggering due to metrics. Why?
A: CloudWatch metric granularity insufficient, metric delay.
238. Q: Scenario: Monitoring multiple instances with same app. Solution?
A: Tag instances → CloudWatch metric filters → dashboards per environment.
239. Q: Scenario: Alert only if sustained high CPU. How?
A: Set **evaluation periods** and threshold accordingly.
240. Q: Scenario: Need to monitor ephemeral storage. How?
A: CloudWatch Agent → disk metrics → create alarm.
241. Q: CloudWatch Events vs CloudWatch Alarms?
A: Events = schedule/trigger actions; Alarms = notify on thresholds.

242. Q: Scenario: Multiple apps on same EC2. Monitor separately?

A: Use **custom namespace metrics** in CloudWatch.

243. Q: Scenario: EC2 network spikes. How to investigate?

A: Check CloudWatch NetworkIn/Out, packet loss, instance type limits.

244. Q: Scenario: Sudden memory spikes. Solution?

A: CloudWatch custom metrics → trigger alarm → auto-restart or scale.

245. Q: EC2 disk burst metrics?

A: BurstBalance metric for gp2 volumes.

246. Q: Scenario: EBS IOPS throttled. Fix?

A: Upgrade to higher provisioned IOPS or gp3 volume.

247. Q: Scenario: EC2 monitoring cost too high. Solution?

A: Consolidate metrics, remove unnecessary detailed monitoring, archive logs.

248. Q: Scenario: Track application errors across EC2 fleet. How?

A: CloudWatch Logs group + Metric Filters + Alarms.

249. Q: Scenario: EC2 instance unresponsive, metrics normal. Why?

A: OS-level deadlock, app crash; check logs, X-Ray, SSM session.

250. Q: Scenario: Historical analysis of EC2 CPU. Solution?

A: Export CloudWatch metrics to S3 → analyze via Athena/QuickSight.

251–300: Cost Optimization & Pricing

251. Q: EC2 pricing models?

A: On-demand, Reserved, Spot, Savings Plans, Dedicated Hosts.

252. Q: Scenario: Dev environment, stop/start frequently. Which model?

A: On-demand for flexibility or spot if interruptions acceptable.

253. Q: Scenario: Long-running production app. Cost-effective?

A: Reserved Instances or Compute Savings Plan.

254. Q: What is Spot instance?

A: Spare capacity at discount; can be interrupted.

255. Q: Scenario: Spot instance terminated mid-workload. Solution?

A: Use **Spot Fleet with diversifications** or checkpointing in app.

256. Q: Difference between RI Standard and Convertible?

A: Standard = fixed instance type, cheaper; Convertible = flexible instance type, slightly higher cost.

257. Q: Scenario: Budget alert for EC2 overspend. How?

A: AWS Budgets → Cost Alerts → Email/SNS notifications.

258. Q: How to reduce EBS cost?

A: Delete unused volumes, snapshots; use gp3 instead of gp2; resize volumes.

259. Q: Scenario: Auto Scaling creates too many instances at night. Solution?

A: Use scheduled scaling policies or limit max capacity.

260. Q: Spot + On-demand hybrid. Scenario?

A: Use On-demand for base capacity, Spot for bursts.

261–300: Cost Optimization & Pricing (missing)

261. Q: Scenario: Production instance running 24/7, cost high. How?

A: Use **Reserved Instances** or Compute Savings Plan.

262. Q: Scenario: Spot instance terminated mid-task. Solution?

A: Use **Spot Fleet** or checkpointing.

263. Q: Scenario: Multiple dev/test instances idle. How to reduce cost?

A: Schedule stop/start via Lambda or Instance Scheduler.

264. Q: Scenario: Long-term dev instances. Cost-effective?

A: Reserved Instances for dev environments.

265. Q: Scenario: EC2 + EBS underutilized. Cost saving?

A: Right-size instance, delete unused volumes, snapshot lifecycle.

266. Q: Spot + On-demand hybrid. Use case?

A: Base load on On-demand, burst via Spot for cost savings.

267. Q: Scenario: Need flexible instance type. Solution?

A: Convertible Reserved Instances.

268. Q: Scenario: Budget alerts for EC2. How?

A: AWS Budgets → set threshold → SNS notifications.

269. Q: Scenario: EBS snapshot costs high. Fix?

A: Enable **lifecycle policies**, delete older snapshots.

270. Q: EC2 pricing options?

A: On-demand, Reserved, Spot, Savings Plans, Dedicated Hosts.

271. Q: Scenario: Spot instance frequently interrupted. Solution?

A: Diversify instance types and AZs in Spot Fleet.

272. Q: Scenario: Minimize cost for dev cluster. Solution?

A: Use **Spot**, schedule off hours, delete unused resources.

273. Q: Scenario: Cost monitoring for multiple accounts. How?

A: Use **AWS Organizations + Cost Explorer**.

274. Q: Scenario: EC2 cost spike. How to analyze?

A: CloudWatch metrics + Cost Explorer + Tags analysis.

275. Q: Scenario: Long-running workloads with predictable traffic. Optimal model?

A: Reserved Instances or Savings Plan.

276. Q: Scenario: Temporary batch workloads. Cost-effective?

A: Spot instances or On-demand depending on tolerance.

277. Q: Scenario: EBS volumes idle. Cost saving?

A: Snapshot and delete unused volumes.

278. Q: Scenario: Multi-AZ ASG scaling increases cost. Solution?

A: Scheduled scaling, use Spot for extra capacity.

279. Q: Scenario: Cross-region backup cost high. Fix?

A: Use S3 lifecycle policies, compress snapshots.

280. Q: Scenario: Spot instance lost, job critical. Solution?

A: Maintain base On-demand instance, use Spot for burst.

281. Q: Scenario: Reserved instance underutilized. How to fix?

A: Modify RI or use **Convertible RI**.

282. Q: Scenario: Need cost alert per project. How?

A: Use tags + Cost Explorer filters + Budgets.

283. Q: Scenario: Large fleet, want automated rightsizing. How?

A: Use **AWS Compute Optimizer** recommendations.

284. Q: Scenario: Multiple instance types in ASG. Cost-optimal?

A: Use **Mixed Instances Policy** with cost-optimized weighting.

285. Q: Scenario: Reducing dev/test spend. How?

A: Stop instances at night, Spot instances, consolidated billing.

286. Q: Scenario: EC2 + EBS combination, cost analysis. How?

A: Use **Cost Explorer**, filter by resource type.

287. Q: Scenario: High IO workloads expensive. Optimize?

A: Use gp3 volume type, provisioned IOPS carefully.

288. Q: Scenario: Budget alert delay. How to ensure near real-time?
A: Use **CloudWatch billing metrics** + SNS notifications.
289. Q: Scenario: Temporary project in multiple regions. How to reduce cost?
A: Use Spot, limit multi-region replication, delete resources post-project.
290. Q: Scenario: EBS snapshot lifecycle policies. Benefits?
A: Automates deletion, reduces cost, maintains compliance.
291. Q: Scenario: Long-running dev instance using On-demand. Alternative?
A: Reserved Instance or Savings Plan.
292. Q: Scenario: Cost optimization for ASG with mixed workloads.
A: Spot for flexible workloads, On-demand for critical base load.
293. Q: Scenario: EBS throughput optimized for batch jobs. Cost-effective?
A: Use st1 (Throughput Optimized HDD) instead of SSD.
294. Q: Scenario: EC2 monitoring costs high. Solution?
A: Remove unnecessary detailed metrics, aggregate metrics.
295. Q: Scenario: Frequent snapshot creation cost. Fix?
A: Use lifecycle manager with retention policies.
296. Q: Scenario: Cross-account EC2 cost visibility. How?
A: Use **AWS Organizations + Cost Explorer**.
297. Q: Scenario: High CPU instances. Right-sizing approach?
A: Use smaller instance with optimized CPU or Reserved for steady load.
298. Q: Scenario: Spot interruption affects SLA. Mitigation?
A: Maintain base On-demand instances, checkpoint workloads.
299. Q: Scenario: Multi-region deployment for HA, cost optimization?
A: Use minimal required instances in secondary region; replicate only critical data.
300. Q: Scenario: EBS IO bottleneck causing extra costs. Solution?
A: Optimize volume type, consider gp3 with provisioned IOPS, monitor usage.

301–350: High Availability & Fault Tolerance

301. Q: Scenario: Single AZ fails, app downtime. Solution?
A: Multi-AZ deployment with ELB and ASG.
302. Q: Difference between Multi-AZ and Multi-Region?

**

A:** Multi-AZ = within region; Multi-Region = geographically separate.

303. Q: Scenario: EC2 instance fails due to hardware. Next step?

A: Stop/start instance or ASG replaces unhealthy instance.

304. Q: How to achieve cross-region failover?

A: Replicate AMIs, volumes, use Route53 failover policies.

305. Q: Scenario: Need ultra-low-latency between EC2s. Solution?

A: Cluster placement group.

306. Q: How to handle AZ failures for databases?

A: Use RDS Multi-AZ or DB replication.

307. Q: Difference between spread and cluster placement group?

A: Spread = isolate for HA; Cluster = low-latency/high throughput.

308. Q: Scenario: EC2 instance becomes unresponsive. How to recover?

A: Check system status, snapshot volume, redeploy instance.

309. Q: How to implement backup for EC2?

A: EBS snapshots, AMI creation, S3 storage.

310. Q: Scenario: Frequent AMI backups cause costs. Solution?

A: Use lifecycle policies to delete old AMIs/snapshots.

311–350: High Availability & Fault Tolerance (missing)

311. Q: Scenario: Single AZ fails, app goes down. How to prevent?

A: Deploy EC2 instances across multiple AZs with **ASG + ELB** for failover.

312. Q: Scenario: Instance fails due to hardware. How to recover?

A: Stop/start instance or rely on ASG health checks to replace.

313. Q: EC2 high availability best practice?

A: Multi-AZ deployment, ELB distribution, health checks, ASG.

314. Q: Scenario: Database instance needs HA. Solution?

A: Use RDS Multi-AZ or EC2 + replication setup across AZs.

315. Q: Scenario: Ultra-low-latency instance communication needed. Solution?

A: Cluster placement group for low-latency network.

316. Q: Difference between cluster and spread placement groups?

A: Cluster = low-latency/high-throughput; Spread = fault isolation.

317. Q: Scenario: EC2 unresponsive. How to troubleshoot?
A: Check CloudWatch metrics, system logs, SSM session, network.
318. Q: Scenario: Backup required before patching. How?
A: Take EBS snapshot or AMI for recovery.
319. Q: Scenario: Restore failed instance quickly. Solution?
A: Launch from latest AMI snapshot.
320. Q: Scenario: EBS root volume corrupted. Recovery?
A: Detach, attach to rescue instance, repair, reattach.
321. Q: EC2 multi-AZ load balancing. How?
A: Deploy in multiple AZs + ELB; ASG ensures desired capacity.
322. Q: Scenario: Instance in one AZ cannot communicate with another. Fix?
A: Verify VPC/subnet routing, SGs, NACLs.
323. Q: Scenario: EC2 instance needs cross-region replication. Solution?
A: Copy AMI or snapshot to target region, launch new instance.
324. Q: Scenario: AZ fails, traffic redirected. How?
A: ELB detects healthy instances in other AZs; Route53 can be used for failover.
325. Q: Scenario: Single instance critical, want auto-recovery. Solution?
A: Enable **EC2 Auto-Recovery** via CloudWatch Alarm.
326. Q: Scenario: Placement group limits reached. Fix?
A: Launch fewer instances or use multiple placement groups.
327. Q: Scenario: Multi-AZ deployment, AZ capacity issue. Solution?
A: Launch in alternative AZ or use ASG flexibility.
328. Q: Scenario: High availability testing. Approach?
A: Simulate AZ failure, validate failover, monitor metrics.
329. Q: Scenario: EC2 system status check fails repeatedly. Solution?
A: Stop/start instance; contact AWS if hardware failure persists.
330. Q: Scenario: Application requires synchronous replication across AZs. Solution?
A: Use database replication or shared EFS for file sync.
331. Q: Scenario: Instance store needed for high-speed cache. How to manage HA?
A: Use **ASG + ephemeral storage**, recover on instance replacement.
332. Q: Scenario: ELB health check removes healthy instance. Fix?
A: Verify app port, SG rules, health check path/response codes.

333. Q: Scenario: Multi-AZ database read replication. Approach?
A: Use read replicas (RDS or self-managed replication).
334. Q: Scenario: Backup strategy for HA EC2. How?
A: EBS snapshots, AMI, cross-AZ or cross-region copies.
335. Q: Scenario: EC2 cluster placement required. Which instances?
A: Instances supporting placement groups (e.g., C5, M5).
336. Q: Scenario: High network throughput bottleneck. Fix?
A: Enable **Enhanced Networking**, select higher network instance type.
337. Q: Scenario: EC2 loses connectivity after reboot. Fix?
A: Check ENI attachment, private/public IP, SG/NACL rules.
338. Q: Scenario: Multi-AZ failover for database. How to minimize downtime?
A: Pre-warm standby, synchronous replication, automated failover scripts.
339. Q: Scenario: Application state lost on failover. Solution?
A: Use external session store (Redis, DynamoDB) or shared storage.
340. Q: EC2 instance health check vs system status check?
A: Instance = application/OS health; system = underlying hardware.
341. Q: Scenario: ASG replacing instances too slowly. How to improve?
A: Reduce cooldown, check launch template, scale-out policy.
342. Q: Scenario: Instance terminated accidentally. How to recover?
A: Restore from AMI snapshot.
343. Q: Scenario: Temporary high throughput workload. Solution?
A: Use Spot or burstable instances + placement group.
344. Q: Scenario: Multi-AZ deployment costs high. Optimization?
A: Use minimal instances in secondary AZ for standby, scale up only when needed.
345. Q: Scenario: Instance stuck pending. Fix?
A: Try alternative AZ or instance type.
346. Q: Scenario: Instance terminated during ASG scale-in. Prevention?
A: Lifecycle hook to drain connections before termination.
347. Q: Scenario: EBS Multi-Attach for shared database. Limitation?
A: Supported only for io1/io2 volumes, Linux instances, same AZ.
348. Q: Scenario: Application requires synchronous replication across regions. Solution?
A: Use database-level replication; Route53 latency-based routing.

349. Q: Scenario: Instance unreachable post-patch. Fix?

A: Check security group, OS firewall, service logs, revert patch if needed.

350. Q: Scenario: EC2 cluster HA testing. Steps?

A: Simulate instance/AZ failures, verify failover, monitor metrics, validate SLA.

351–400: DevOps & Automation

351. Q: How to automate EC2 provisioning?

A: Use CloudFormation, Terraform, or CDK scripts.

352. Q: Scenario: Hundreds of instances need same config. How?

A: Use **Launch Templates** or automation scripts (Terraform/CloudFormation).

353. Q: How to configure EC2 using scripts?

A: Use **User Data** during launch or Systems Manager Run Command.

354. Q: Scenario: Deploy app updates to multiple instances. How?

A: Use **SSM Run Command** or **CodeDeploy**.

355. Q: Can Terraform manage EC2?

A: Yes, with `aws_instance`, `aws_launch_template`, `aws_autoscaling_group`.

356. Q: Scenario: Need versioned EC2 deployment. How?

A: Use **Launch Template versions**.

357. Q: How to integrate CI/CD with EC2?

A: Use CodePipeline/CodeDeploy to push code to EC2 instances.

358. Q: Scenario: App config differs in prod vs dev. Solution?

A: Use **parameter store** or **Secrets Manager** for environment-specific configs.

359. Q: How to run commands on EC2 without SSH?

A: Use **SSM Session Manager**.

360. Q: Scenario: EC2 behind firewall, no internet. Remote commands needed. Solution?

A: SSM with VPC endpoints for SSM service.

361–400: DevOps & Automation / Real-world Outages

361. Q: Scenario: Hundreds of EC2s need same config. Solution?

A: Use **Terraform** or **CloudFormation** templates.

362. Q: Scenario: App updates to multiple instances. Solution?

A: Use **SSM Run Command** or **CodeDeploy**.

363. Q: Scenario: Instance in private subnet needs remote commands. How?

A: Use **SSM Session Manager** with VPC endpoints.

364. Q: Scenario: CI/CD pipeline for EC2. How?

A: CodePipeline + CodeDeploy to deploy apps to instances.

365. Q: Scenario: User Data script fails on launch. Fix?

A: Verify syntax, logs at /var/log/cloud-init.log.

366. Q: Scenario: Terraform destroys instance by mistake. Mitigation?

A: Use lifecycle { prevent_destroy = true } in config.

367. Q: Scenario: EC2 configuration drift. Solution?

A: Use Systems Manager State Manager for compliance.

368. Q: Scenario: Auto-remediation for unhealthy EC2. How?

A: CloudWatch Alarm → Lambda → SSM → restart instance.

369. Q: Scenario: Instance compromised. Immediate action?

A: Isolate, snapshot, investigate, rotate keys, redeploy.

370. Q: Scenario: Spot instance interruption. Mitigation?

A: Use **Spot Fleet** diversification, checkpoint workload.

371. Q: Scenario: Application crashed silently. How to detect?

A: CloudWatch Logs, X-Ray tracing, custom alarms.

372. Q: Scenario: EC2 batch job causes CPU spike. How to handle?

A: Scale via ASG, schedule batch jobs, monitor CloudWatch metrics.

373. Q: Scenario: EC2 fails system status check. Fix?

A: Stop/start instance; AWS may replace underlying hardware.

374. Q: Scenario: Network latency spikes between EC2s. Fix?

A: Check placement groups, instance type network throughput, cross-AZ traffic.

375. Q: Scenario: EC2 cannot mount EFS. Fix?

A: Security group, subnet routing, mount target, NFS client config.

376. Q: Scenario: Disk full on EC2. Mitigation?

A: CloudWatch disk usage metrics, clean logs, extend EBS volume.

377. Q: Scenario: Auto-scaling fails during peak. Fix?

A: Check CloudWatch metrics, scaling policy, cooldown, MaxSize.

378. **Q:** Scenario: Multiple EC2s need central logging. How?
A: CloudWatch Logs centralized group, log agent installed.
379. **Q:** Scenario: Application memory leak. How to monitor?
A: CloudWatch custom metrics, agent installed, alarms.
380. **Q:** Scenario: Cross-region latency affects app. Solution?
A: Use Route53 latency-based routing, deploy app closer to users.
381. **Q:** Scenario: EC2 patching needs automation. How?
A: Systems Manager Patch Manager, maintenance window.
382. **Q:** Scenario: App requires zero-downtime deployment. Solution?
A: Blue/Green deployment using ASG and ELB.
383. **Q:** Scenario: Multi-AZ EC2 fails partially. Recovery?
A: ASG launches new instance in healthy AZ.
384. **Q:** Scenario: Instance metadata attack prevention.
A: IMDSv2 enforced, IAM role restriction, patch OS.
385. **Q:** Scenario: EC2 requires audit trail. Solution?
A: CloudTrail logs, Config rules, CloudWatch Events.
386. **Q:** Scenario: EC2 cost spike detected. Response?
A: CloudWatch + Budget alarms, analyze via Cost Explorer.
387. **Q:** Scenario: Spot instance interrupted during critical processing. Solution?
A: Maintain base On-demand instance + checkpoint workload.
388. **Q:** Scenario: Auto Scaling replaces healthy instance repeatedly. Fix?
A: Verify termination policies, health check configuration.
389. **Q:** Scenario: EC2 instance stuck pending. Solution?
A: AZ capacity issue → try different AZ or instance type.
390. **Q:** Scenario: EC2 cannot reach database. Fix?
A: Check SG, route tables, VPC peering, DNS resolution.
391. **Q:** Scenario: Lifecycle hook for ASG scale-in. Use?
A: Pause termination to drain connections or backup data.
392. **Q:** Scenario: EC2 network packets dropped. How to investigate?
A: CloudWatch metrics, ENI stats, instance type limits.
393. **Q:** Scenario: ELB removes instances too aggressively. Fix?
A: Adjust health check intervals and thresholds.

394. **Q:** Scenario: EC2 requires sub-second failover. Solution?
A: Pre-warmed standby, cluster placement group, ASG readiness.
395. **Q:** Scenario: High-availability testing for EC2. Steps?
A: Simulate instance/AZ failure, validate failover, monitor SLA metrics.
396. **Q:** Scenario: EC2 storage performance throttled. Fix?
A: Upgrade volume type (gp3/io2), increase IOPS, enable EBS optimization.
397. **Q:** Scenario: Cross-region DR test for EC2. How?
A: Replicate AMI/snapshots, launch test instances, validate failover.
398. **Q:** Scenario: EC2 audit compliance. How?
A: Enable CloudTrail, Config, Security Hub, patching compliance.
399. **Q:** Scenario: EC2 application logs need centralized search. Solution?
A: CloudWatch Logs → Export to S3 → Athena/Elasticsearch for query.
400. **Q:** Scenario: EC2 scaling policies not behaving as expected. How?
A: Verify CloudWatch metrics, policy thresholds, evaluation periods, cooldowns.

401–450: Real-World Outages & Troubleshooting

401. **Q:** Scenario: EC2 not reachable after patch. Check?
A: Verify network, security groups, firewall, patch logs.
402. **Q:** EC2 CPU spike during batch jobs. Fix?
A: Use larger instance type, optimize batch jobs, schedule off-peak.
403. **Q:** EC2 disk full. Scenario?
A: Monitor with CloudWatch, extend volume, remove logs.
404. **Q:** Scenario: App fails on instance restart. Cause?
A: Missing User Data script persistence, EBS volume detached.
405. **Q:** Network latency high. Scenario?
A: Check placement group, instance type network throughput, cross-AZ traffic.
406. **Q:** EC2 fails health checks. Solution?
A: Review app logs, security group, ELB config.
407. **Q:** Spot instance interrupted mid-processing. Solution?
A: Checkpointing, use Spot Fleet diversification.
408. **Q:** Scenario: EC2 loses internet in private subnet. Fix?
A: Check NAT gateway, route tables, NACLs.

409. Q: System status check failed on EC2. Solution?

A: Stop/start instance, contact AWS if persists.

410. Q: Instance terminated accidentally. How to recover?

A: Restore from latest AMI or snapshot.

410–450: Spot/Reserved Instances, SRE Practices, Compliance

410. Q: Scenario: Spot instance terminated mid-job. How to prevent data loss?

A: Use **checkpointing**, maintain base On-demand instance, or Spot Fleet with diversified instance types.

411. Q: Scenario: Reserved Instance underutilized. How to optimize?

A: Modify RI type, convert to Convertible RI, or sell unused RI in AWS Marketplace.

412. Q: Scenario: Multi-region deployment for disaster recovery. Cost-effective approach?

A: Keep minimal standby instances in secondary region, replicate only critical data.

413. Q: Scenario: Spot Fleet failing to launch instances. Check?

A: Validate instance types, AZ capacity, allocation strategy, IAM permissions.

414. Q: Scenario: EC2 cost optimization with mixed workloads. How?

A: On-demand for base load, Spot for burst; monitor metrics via CloudWatch.

415. Q: Scenario: Auto-scaling uses Spot, SLA critical. Mitigation?

A: Maintain minimum On-demand instances, checkpoint Spot workloads.

416. Q: Scenario: EC2 compliance requirements for PCI-DSS. How?

A: Encrypt volumes, use IAM roles, audit via Config, enable CloudTrail.

417. Q: Scenario: EC2 running sensitive workloads. Security best practice?

A: Patch OS, restrict IAM roles, enable Security Groups, VPC isolation.

418. Q: Scenario: EC2 cross-account access required. Secure method?

A: Use IAM roles with trust policy; avoid sharing keys.

419. Q: Scenario: EC2 fails patch compliance check. How?

A: Use Systems Manager Patch Manager, remediate automatically.

420. Q: Scenario: Spot instance price fluctuation affects workloads. Solution?

A: Set maximum price, diversify AZs, monitor via CloudWatch.

421. Q: Scenario: SRE wants error budget alert on EC2 application. How?

A: Monitor CloudWatch custom metrics → set alarm → notify via SNS.

422. **Q:** Scenario: EC2 latency spikes for user traffic. How to debug?
 A: Check placement group, network throughput, instance type, CloudWatch metrics.
423. **Q:** Scenario: Multi-AZ deployment fails partially. SRE mitigation?
 A: ASG launches replacement in healthy AZ, monitor SLA.
424. **Q:** Scenario: EC2 application needs zero-downtime deployment. Approach?
 A: Blue/Green deployment using ASG + ELB; gradually shift traffic.
425. **Q:** Scenario: Compliance requires immutable EC2 logs. Solution?
 A: Forward logs to **S3 with versioning + encryption**, restrict write permissions.
426. **Q:** Scenario: EC2 instance reboot causes downtime. Solution?
 A: Enable **Auto-Recovery**, use Multi-AZ deployment, pre-warm standby.
427. **Q:** Scenario: Spot instance terminated, SLA critical. Mitigation?
 A: Use checkpointing, Spot Fleet diversification, maintain minimum On-demand.
428. **Q:** Scenario: EC2 requires audit trail of user access. Solution?
 A: Enable CloudTrail logging, enforce IAM policies, review regularly.
429. **Q:** Scenario: Reserved instance not applied automatically. Why?
 A: Check instance type, region, tenancy, platform match for RI coverage.
430. **Q:** Scenario: Spot instance price increases beyond budget. How?
 A: Auto-terminate or switch to alternative instance type/AZ.
431. **Q:** Scenario: EC2 instances drift from configuration baseline. Solution?
 A: Use Systems Manager State Manager for configuration enforcement.
432. **Q:** Scenario: EC2 network performance drops. Fix?
 A: Enable enhanced networking, check ENI, upgrade instance type.
433. **Q:** Scenario: Multi-AZ failover takes too long. How to improve?
 A: Pre-warm standby, health checks optimized, reduce ASG cooldowns.
434. **Q:** Scenario: EC2 encryption required for compliance. How?
 A: Enable EBS encryption, use KMS keys, encrypt snapshots and AMIs.
435. **Q:** Scenario: EC2 audit report needed weekly. How?
 A: Use AWS Config rules, schedule reports to S3.
436. **Q:** Scenario: Spot instance interruptions causing job failures. How?
 A: Checkpoint, maintain On-demand base, diversify Spot instance types.
437. **Q:** Scenario: Reserved Instance cost savings not realized. Why?
 A: Wrong instance type, region, OS, tenancy mismatch with RI.

438. Q: Scenario: EC2 patching automated for compliance. How?

A: Systems Manager Patch Manager + maintenance windows.

439. Q: Scenario: EC2 instance logs tampered. Mitigation?

A: Forward to immutable storage (S3 with versioning + encryption), monitor via CloudTrail.

440. Q: Scenario: Spot Fleet not maintaining capacity. Fix?

A: Check allocation strategy, diversify AZs and instance types.

441. Q: Scenario: Multi-region EC2 deployment for DR. Cost-efficient setup?

A: Keep minimal standby instances, replicate essential data only.

442. Q: Scenario: EC2 health check fails repeatedly. How?

A: Check OS, application, SG, NACL, CloudWatch metrics.

443. Q: Scenario: Auto Scaling replacing healthy instances. Cause?

A: Misconfigured health checks or termination policies.

444. Q: Scenario: EC2 compliance for HIPAA workloads. Actions?

A: Encrypt data at rest/in transit, IAM restriction, logging, audit trail.

445. Q: Scenario: Spot instance suddenly terminated mid-critical workload. Recovery?

A: Checkpoint state, maintain base On-demand instance, automated replacement.

446. Q: Scenario: EC2 application logs centralized for compliance. How?

A: CloudWatch Logs → S3 → Athena or Elasticsearch for query/audit.

447. Q: Scenario: Reserved Instance modification needed. How?

A: Use AWS console/CLI → modify instance family, AZ, OS, tenancy if Convertible RI.

448. Q: Scenario: EC2 instances need automated scaling + cost optimization. Solution?

A: ASG + Spot Fleet diversification + CloudWatch metric-based policies.

449. Q: Scenario: EC2 SLA not met during AZ failure. Mitigation?

A: Multi-AZ deployment, pre-warmed standby, ASG scaling policies.

450. Q: Scenario: EC2 spot interruptions impacting SRE error budget. How?

A: Use checkpointing, diversify Spot types, maintain On-demand base, alert via CloudWatch + SNS.

451–500: SRE Practices, Compliance & Advanced Topics

451. Q: How to implement EC2 observability?

A: CloudWatch metrics, CloudTrail, X-Ray, custom metrics.

452. Q: Scenario: Auto-remediation for unhealthy EC2. How?

A: Lambda triggered via CloudWatch Alarm → restart instance.

453. **Q:** What is SRE approach for EC2 uptime?
A: Define SLO/SLI, use redundancy, monitoring, automated recovery.
454. **Q:** How to enforce compliance on EC2?
A: IAM policies, Config rules, Security Hub, patch compliance.
455. **Q:** Scenario: EC2 needs PCI DSS compliance. How?
A: Harden OS, encryption at rest & in transit, audit logging, VPC isolation.
456. **Q:** How to handle EC2 scale-down gracefully?
A: Use lifecycle hooks in ASG to drain connections before termination.
457. **Q:** Scenario: Multiple ASGs with different scaling metrics. How?
A: Configure target tracking or step policies per ASG.
458. **Q:** EC2 cost optimization tips?
A: Reserved/spot instances, rightsizing, delete unused volumes, schedule stop/start for dev instances.
459. **Q:** Scenario: EC2 needs multi-region backup for compliance. How?
A: Cross-region AMI copy, EBS snapshot replication.
460. **Q:** EC2 patching best practices?
A: Test in dev, automate via Systems Manager Patch Manager, schedule maintenance window.
461. **Q:** EC2 security best practices?
A: IAM roles, least privilege, encryption, VPC isolation, security group restrictions.
462. **Q:** Scenario: Application cannot reach DB in different AZ. Solution?
A: Check route tables, security group, VPC peering.
463. **Q:** EC2 Spot + Reserved hybrid. Scenario?
A: Reserved for base load, Spot for burst workloads.
464. **Q:** Scenario: EC2 disk performance bottleneck. Fix?
A: Use provisioned IOPS, higher instance type, EBS-optimized instances.
465. **Q:** EC2 with high network throughput requirement. Solution?
A: Enhanced networking enabled (ENA), placement group, higher network instance type.
466. **Q:** Scenario: EC2 instance compromised. Steps?
A: Isolate, snapshot, investigate, rotate keys, redeploy.
467. **Q:** What is EC2 instance recovery?
A: Automatically recovers instance on underlying hardware failure without changing instance ID.
468. **Q:** Scenario: Recovery fails. Next step?
A: Stop/start instance, contact AWS Support if hardware issue persists.

469. **Q:** EC2 monitoring for cost anomaly. How?
 A: CloudWatch + Budgets alerts + Cost Explorer.
470. **Q:** Scenario: Auto Scaling not triggering. Check?
 A: CloudWatch metric, policy configuration, cooldown period.
471. **Q:** EC2 lifecycle hooks?
 A: Pause ASG scale-in/out to perform custom actions (drain connections, backup).
472. **Q:** Scenario: Need zero-downtime deployment. Solution?
 A: Blue/Green deployment with ELB and ASG.
473. **Q:** EC2 multi-tenancy compliance issues?
 A: Use dedicated hosts if regulatory requirements demand physical isolation.
474. **Q:** Scenario: EC2 instances need auditing for security changes. How?
 A: CloudTrail logs, Config rules, CloudWatch Events.
475. **Q:** EC2 high availability checklist?
 A: Multi-AZ, ELB, ASG, health checks, snapshots, monitoring.
476. **Q:** Scenario: Application requires sub-second failover. Solution?
 A: Cluster placement group, pre-warmed standby instances.
477. **Q:** EC2 integration with S3 & EFS for shared storage. Scenario?
 A: Mount EFS across EC2; use S3 SDK for object storage.
478. **Q:** Scenario: EBS snapshot restore slow. How?
 A: Use **fast snapshot restore** option.
479. **Q:** How to secure instance metadata?
 A: IMDSv2 enforced, limit role privileges.
480. **Q:** Scenario: EC2 cross-region DR test. Approach?
 A: Replicate AMI/snapshots, deploy test instances, failover validation.
481. **Q:** How to prevent accidental termination?
 A: Enable **termination protection**.
482. **Q:** Scenario: Instance terminated despite protection. Cause?
 A: Force terminate via API, instance was not protected at launch.
483. **Q:** EC2 network troubleshooting steps?
 A: Check ENI, SG, NACL, route tables, VPC endpoints, DNS resolution.
484. **Q:** Scenario: EC2 cannot resolve private domain. Fix?
 A: Check Route53 private hosted zone, VPC association, DHCP option sets.

485. **Q:** EC2 monitoring for SLA compliance. Approach?

A: CloudWatch SLI metrics, alarm on deviation, Auto Recovery.

486. **Q:** Scenario: EC2 load balancer unhealthy threshold too sensitive. Solution?

A: Adjust health check thresholds or intervals.

487. **Q:** EC2 instance cannot mount EFS NFS. Fix?

A: Security group, mount target, subnet route, NFS client configuration.

488. **Q:** Scenario: High cost due to idle EC2. Solution?

A: Stop unused instances, use spot for dev/test, schedule start/stop.

489. **Q:** EC2 audit trail for compliance. Tools?

A: CloudTrail, Config, Security Hub.

490. **Q:** Scenario: EC2 RDS access denied after instance recreation. Solution?

A: IAM role attachment, security group inbound, subnet route.

491. **Q:** EC2 instance CPU credit exhaustion. Scenario?

A: T2/T3 burstable instance; monitor with CloudWatch, consider larger or standard instance.

492. **Q:** Scenario: Auto Scaling group not scaling to max. Fix?

A: Check ASG MaxSize, CloudWatch metrics, scaling policies.

493. **Q:** EC2 spot interruption notice. Response?

A: Use two-minute notice for checkpointing or migrate workload.

494. **Q:** EC2 instance stuck pending. Scenario?

A: Insufficient capacity in AZ; try different AZ or instance type.

495. **Q:** Scenario: EC2 fails after AMI update. Solution?

A: Validate AMI, dependencies, test in staging, rollback if needed.

496. **Q:** EC2 instance fails system status check repeatedly. Approach?

A: Stop/start, check EBS, contact AWS for hardware replacement.

497. **Q:** Scenario: EC2 memory leak in app. Detection?

A: CloudWatch custom metrics, detailed monitoring, logs.

498. **Q:** EC2 instance metadata poisoning attack. Prevention?

A: IMDSv2 enforced, restrict IAM privileges, patch OS.

499. **Q:** Scenario: EC2 fleet experiencing intermittent network failures across AZs. How to troubleshoot and mitigate?

A: Check VPC/subnet routing, SG/NACL rules, placement groups, instance type network limits, monitor CloudWatch metrics, enable enhanced networking, and consider cross-AZ traffic optimization.

500. **Q:** Scenario: Multi-AZ EC2 deployment costs exceeding budget while maintaining HA. Optimization strategy?

A: Keep minimal standby in secondary AZ, use Spot instances for non-critical workloads, rightsizing, enable auto-start/stop for dev/test, monitor costs via AWS Budgets, enforce lifecycle policies for snapshots and AMIs.

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